DO NOW — RECALL

Lesson 069 · Data Mining and Visualisation of Patterns

Name: _____ Class: _____ Date: _____

1. What is a backtracking branch?

2. What is an outlier?

3. What is the purpose of a scatter graph?

4. Array [2,3,4,5]: double elements at indexes 1 and 3. Give updated array and sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

One choice explored by search.

2. Model answer

Value unusually different from others.

3. Model answer

Displays association between two numerical variables.

4. Model answer

[2,6,4,10]; total22.

By the end, you can…

• Data mining seeks patterns or associations across sets of data; define the question, prepare records and examine the results.

• Visualisation makes patterns and anomalies visible. Scatter charts suit pairs of numeric data; bar charts compare categories.

• A correlation does not establish cause. Sampling, confounders, missing values and measurement choices may mislead interpretation.

Vocabulary — know these terms

data mining — Finding patterns in datasets.

scatter plot — Pairs of numeric variables plotted as points.

outlier — Unusually distant observation.

correlation — Association, not proof of causation.

NEW LEARNING

Read, annotate and explain

Define and predict

Data mining seeks patterns or associations across sets of data; define the question, prepare records and examine the results.

Explain

Visualisation makes patterns and anomalies visible. Scatter charts suit pairs of numeric data; bar charts compare categories.

Evaluate

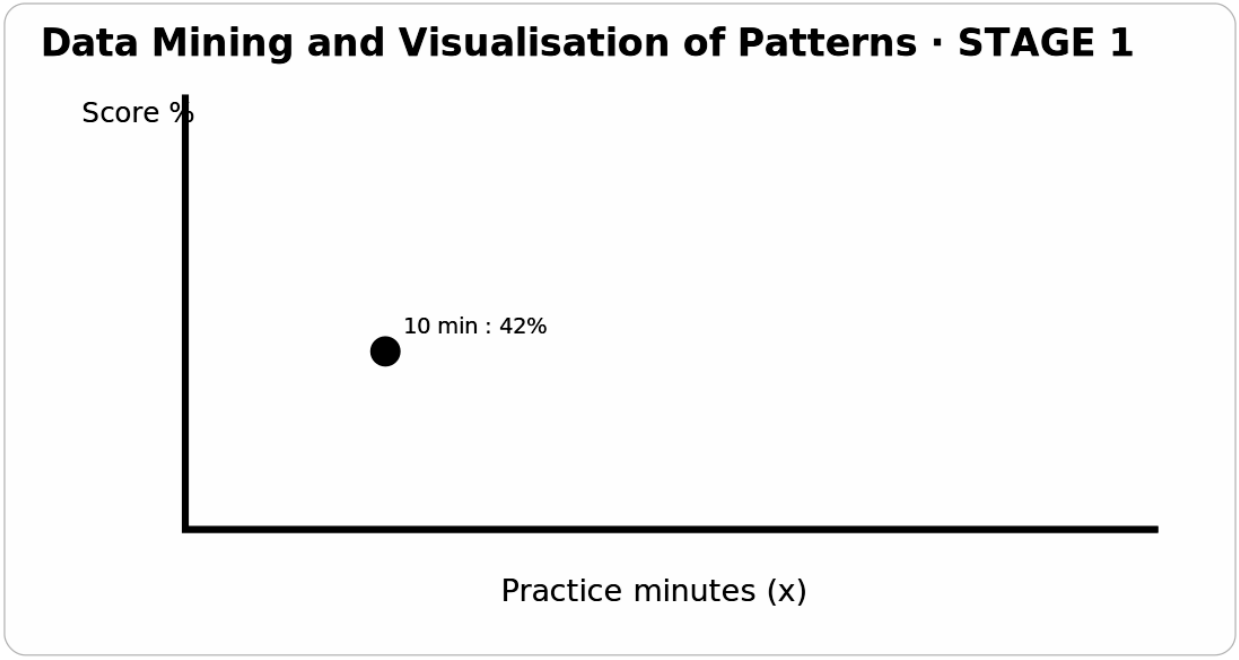
A correlation does not establish cause. Sampling, confounders, missing values and measurement choices may mislead interpretation.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

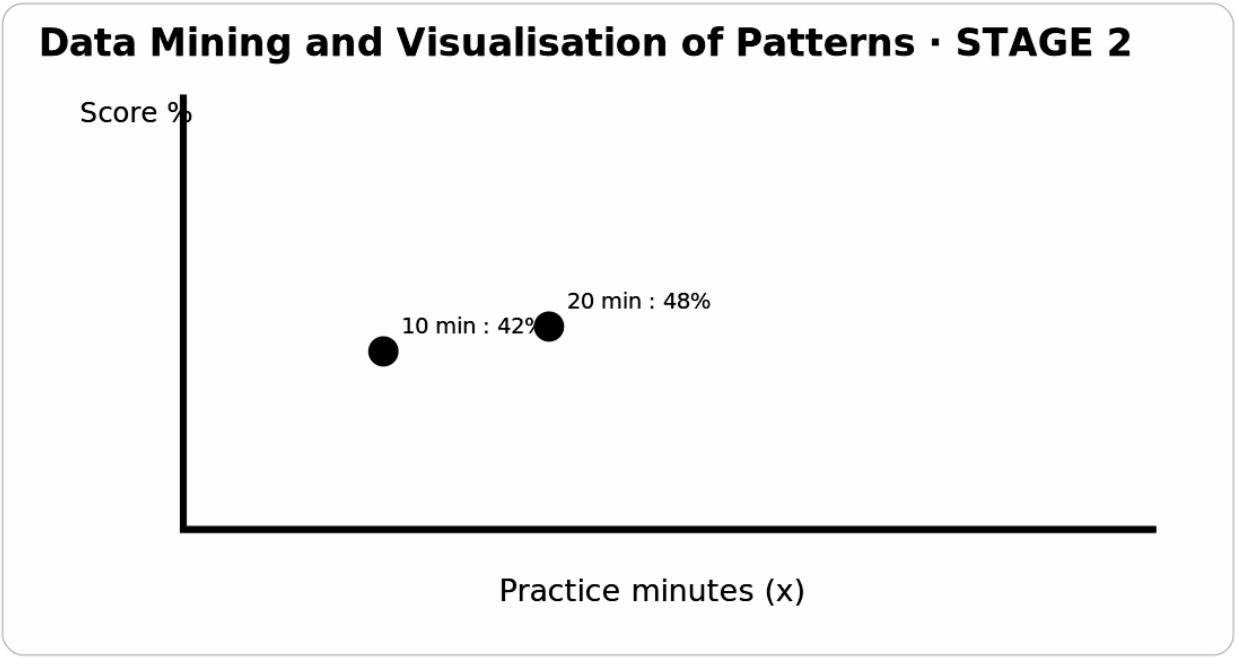


Data Mining and Visualisation of Patterns · STAGE 1; 10 min → 42%

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Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

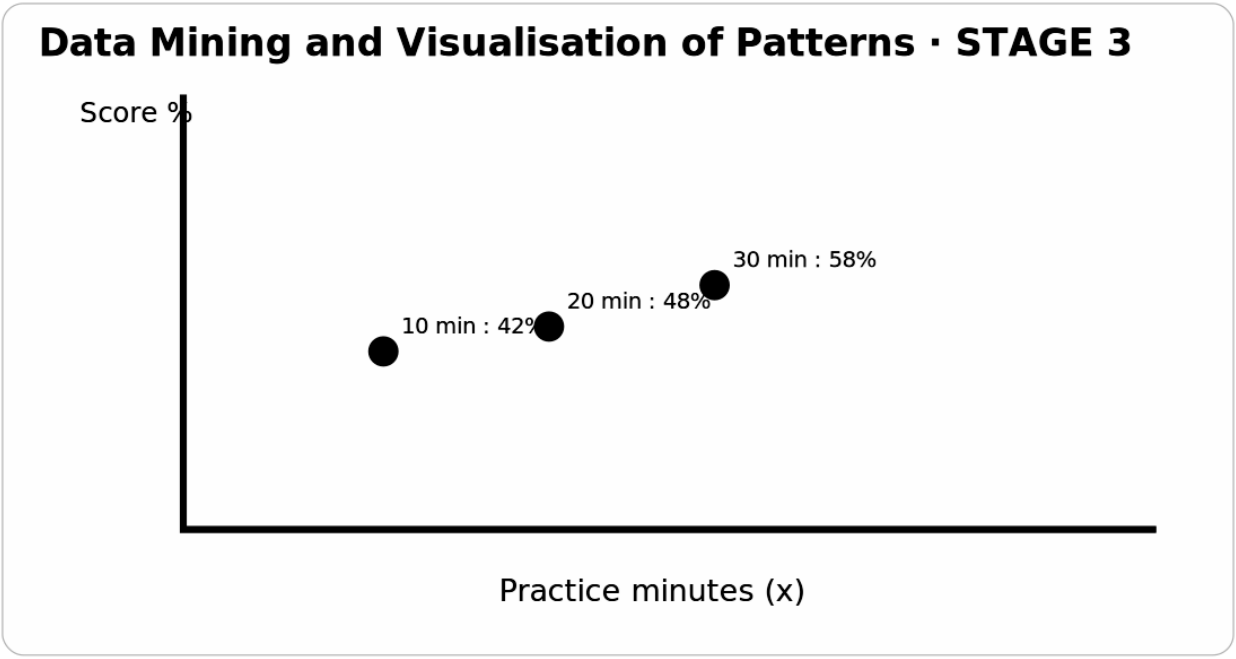


Data Mining and Visualisation of Patterns · STAGE 2; 10 min → 42%; 20 min → 48%

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

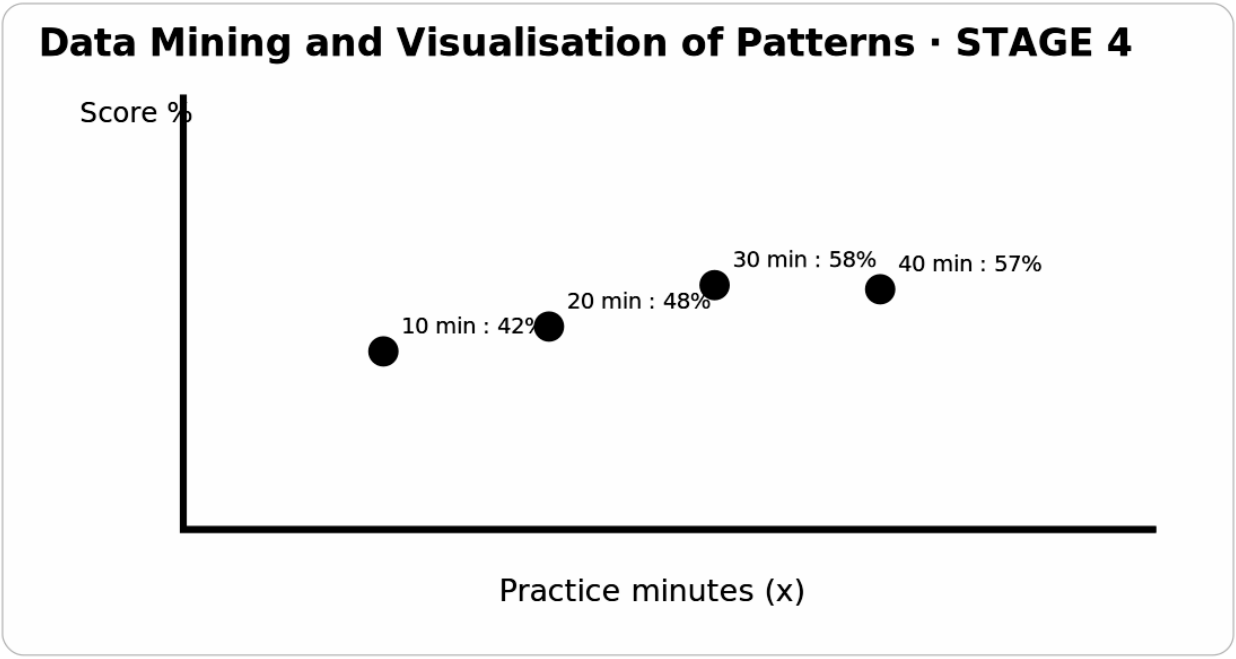


Data Mining and Visualisation of Patterns · STAGE 3; 10 min → 42%; 20 min → 48%; 30 min → 58%

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

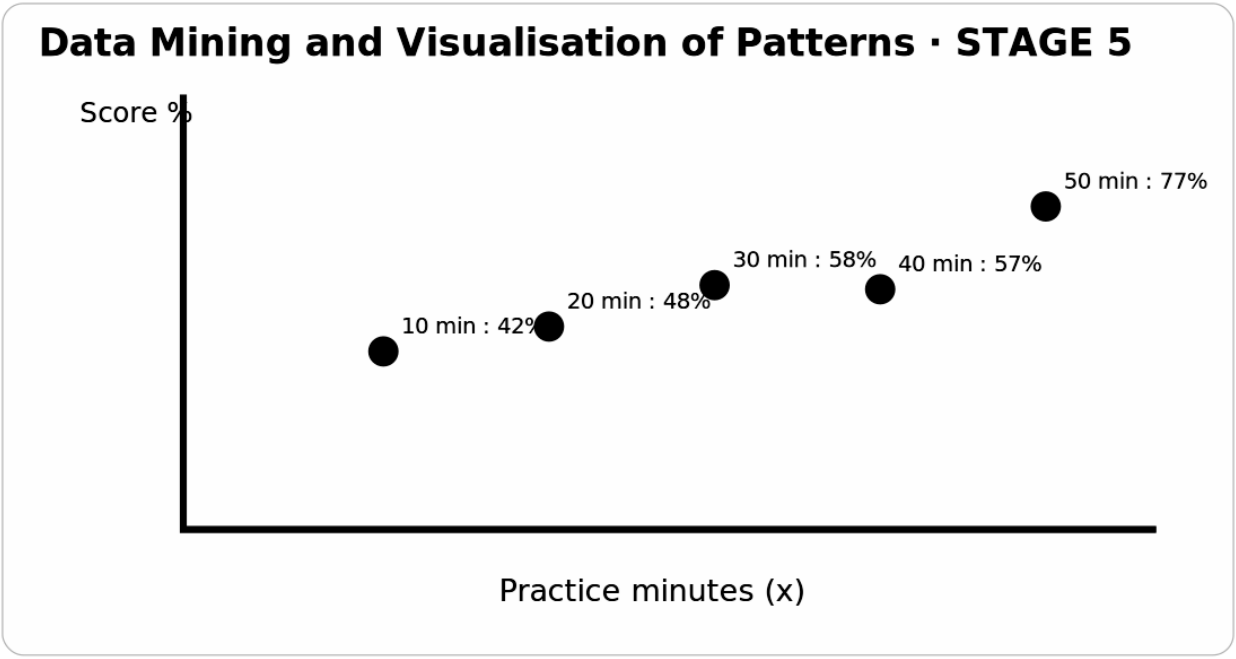


Data Mining and Visualisation of Patterns · STAGE 4; 10 min → 42%; 20 min → 48%; 30 min → 58%; 40 min → 57%

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Data Mining and Visualisation of Patterns · STAGE 5; 10 min → 42%; 20 min → 48%; 30 min → 58%; 40 min → 57%; 50 min → 77%

I DO — WORKED EXAMPLE

Follow the worked method

For points (1,2),(2,4),(3,6), state trend and one limitation.

Follow the reasoning

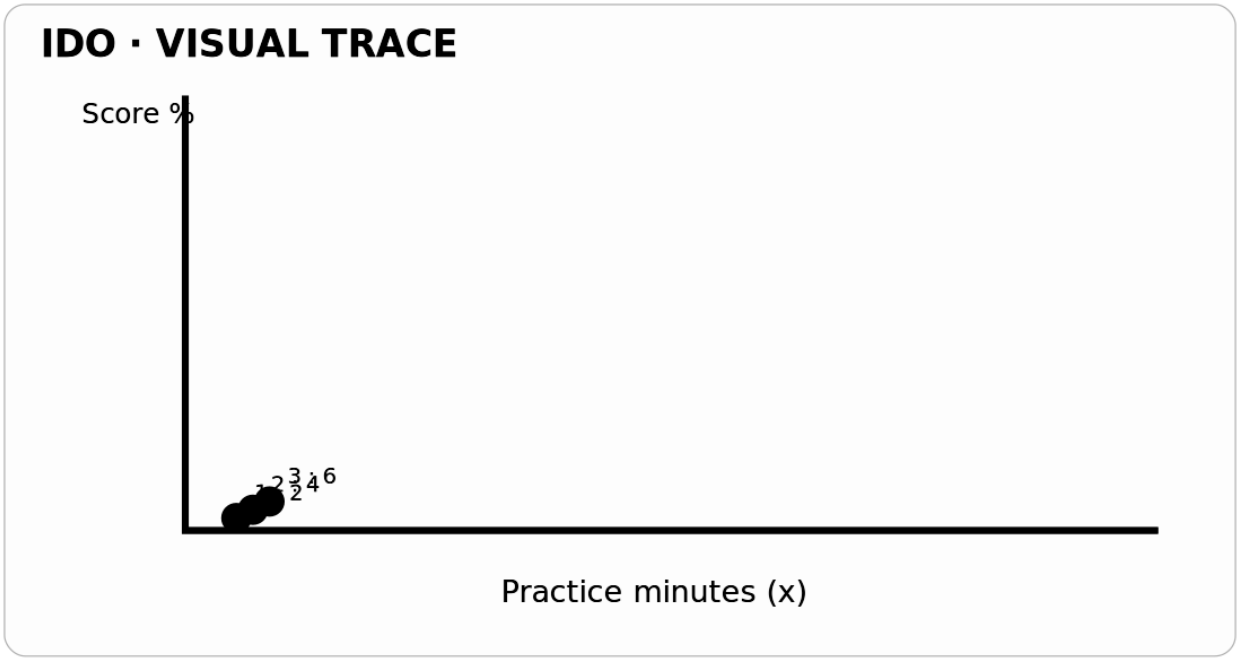
Separate observation from causal explanation.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For points (1,2),(2,4),(3,6), state trend and one limitation.

Separate observation from causal explanation.



IDO · VISUAL TRACE; 1 → 2; 2 → 4; 3 → 6

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

For scores [40,42,46,96], calculate mean and median; explain influence of outlier.
Source code / input:
Mean (40+42+46+96)/4=56; median (42+46)/2=44.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

For points (1,2),(2,4),(3,6), choose a suitable chart and describe association. [4]

Question 2

For [40,42,46,96] calculate mean and median with working. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate mining a school's student-usage logs to predict attainment. Address fairness, data quality and privacy. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A rising scatter trend proves that extra practice alone caused higher marks.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

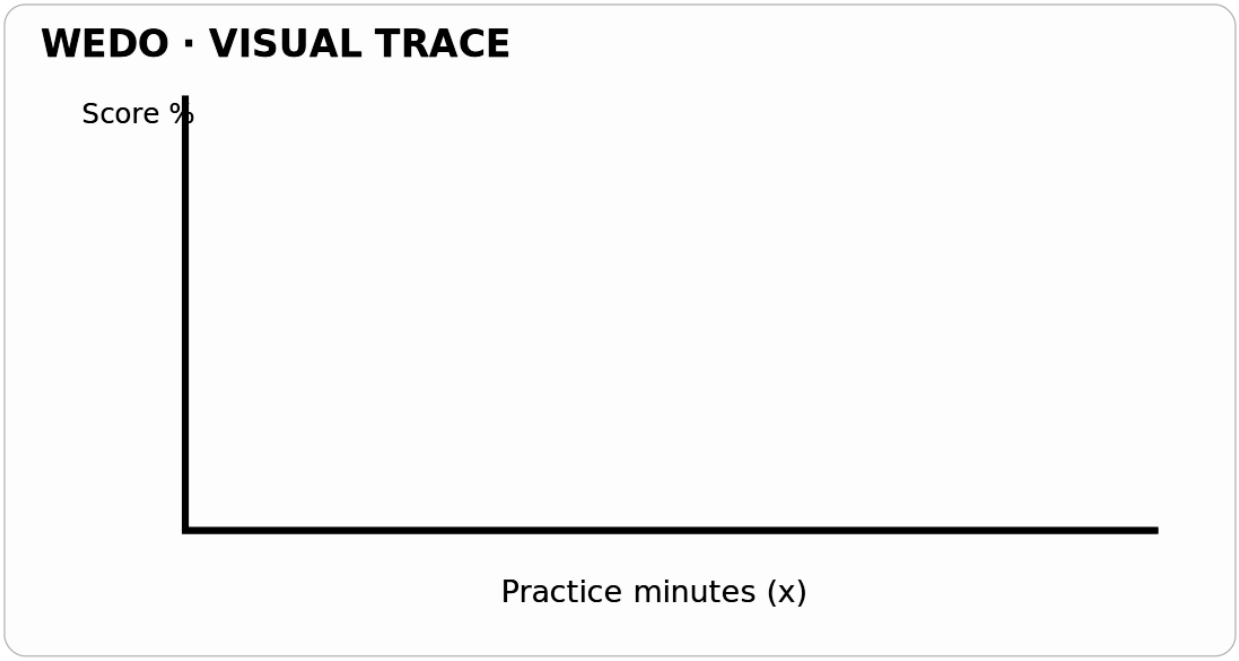
Follow the reasoning
The value96 pulls mean away from three clustered lower scores.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For scores [40,42,46,96], calculate mean and median; explain influence of outlier.

The value96 pulls mean away from three clustered lower scores.



WEDO · VISUAL TRACE; 40; 42; 46; 96